

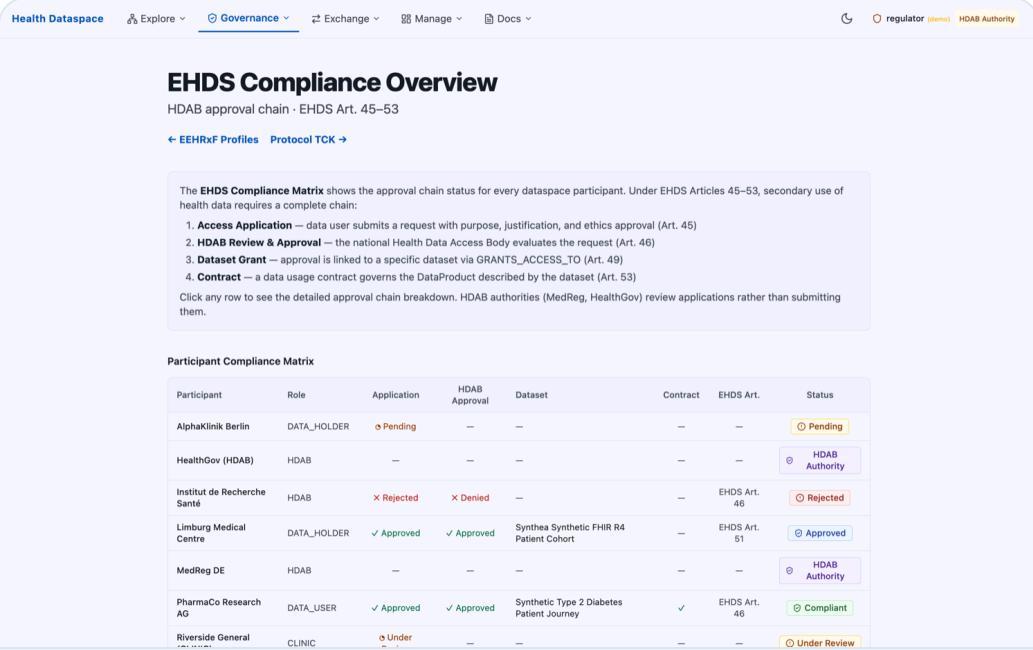
Consent Wallet for the European Health Data Space

Watch my phone, not a slide — the EUDI Wallet as the citizen's verifiable, withdrawable channel to exercise their EHDS secondary-use rights: opt out by default, consent where the law requires. Running today on a real EHDS reference stack — synthetic data only.

“From honeypot to handset: opt out in your wallet, and the research gate goes dark.”

1 Watch my phone, not a slide

- Lena (34, synthetic patient) opts out of research use — in her EUDI Wallet
- Re-run the governance gate → researcher access DENIED
- Synthetic data · fictional orgs · standard eIDAS 2.0 + EHDS protocols
- The idea: consent & opt-out as a wallet-held, verifiable, withdrawable trust primitive — no central registry



EHDS Compliance Overview
HDAB approval chain · EHDS Art. 45–53

← EHRxProfiles Protocol TCK →

The **EHDS Compliance Matrix** shows the approval chain status for every dataspace participant. Under EHDS Articles 45–53, secondary use of health data requires a complete chain:

1. **Access Application** — data user submits a request with purpose, justification, and ethics approval (Art. 45)
2. **HDAB Review & Approval** — the national Health Data Access Body evaluates the request (Art. 46)
3. **Dataset Grant** — approval is linked to a specific dataset via GRANTS_ACCESS_TO (Art. 49)
4. **Contract** — a data usage contract governs the DataProduct described by the dataset (Art. 53)

Click any row to see the detailed approval chain breakdown. HDAB authorities (MedReg, HealthGov) review applications rather than submitting them.

Participant	Role	Application	HDAB Approval	Dataset	Contract	EHDS Art.	Status
AlphaKlinik Berlin	DATA HOLDER	Pending	—	—	—	—	Pending
HealthGov (HDAB)	HDAB	—	—	—	—	—	HDAB Authority
Institut de Recherche Santé	HDAB	Rejected	Denied	—	—	EHDS Art. 46	Rejected
Limburg Medical Centre	DATA HOLDER	Approved	Approved	Synthesa Synthetic FHIR R4 Patient Cohort	—	EHDS Art. 51	Approved
MedReg DE	HDAB	—	—	—	—	—	HDAB Authority
PharmaCo Research AG	DATA_USER	Approved	Approved	Synthetic Type 2 Diabetes Patient Journey	✓	EHDS Art. 46	Compliant
Riverside General	CLINIC	Under Review	—	—	—	—	Under Review

MVHD governance gate — APPROVED, then DENIED after Lena opts out

2 The regulation ships a gap

- EHDS (Reg. 2025/327, Ch. IV): secondary use runs on an HDAB data permit + a citizen right to opt out — no reason needed
- eIDAS 2.0 (Reg. 2024/1183) ships the wallet, PID and EAA — the missing citizen-facing channel
- The gap: no standardised, citizen-held, machine-verifiable opt-out + consent channel
- Default fallback = a central opt-out registry: a honeypot the citizen never really controls

VISUAL

Three article-stamped blocks — EHDS Ch. IV (HDAB data permit + opt-out, no reason) · eIDAS 2.0 (wallet, PID, EAA, selective disclosure) · GDPR (HDAB assesses the Art. 6 basis; Art. 7(3) withdrawal where consent applies) — with a glaring red dashed gap in the middle labelled 'NO standardised citizen-held opt-out + consent channel'. Below: a crossed-out central database cylinder captioned 'opt-out honeypot — single point of failure'.

3 The use case, in one sentence

- The EUDI Wallet becomes the citizen's channel to exercise their EHDS secondary-use rights
- Opt out by default · explicit consent where the law requires it · withdraw at any time
- Keycloak issues a verifiable preference attestation — SD-JWT VC, dc+sd-jwt, ES256
- Selective disclosure: the researcher sees patient_pseudonym, never the name (data minimisation)

VISUAL

The wallet card (deep-navy #0b3d66 'EHDS Patient Identity') with literal claim chips: consent_purpose=secondary-use/research · consent_scope=drug-repurposing-study · patient_pseudonym=.... · role=PATIENT, with family_name greyed/withheld to prove selective disclosure. Arrow labelled 'OpenID4VP' to the governance PASS/FAIL panel. Footnote chip: 'sovereignty by architecture — no central opt-out store'.

4 Live demo: grant → govern → withdraw

- 1 GRANT — Lena gets a PID via the EUDI reference issuer; presents at the MVHD portal; her preference attestation is issued back to the wallet
- 2 GOVERN — MedReg DE (HDAB) issues PharmaCo's data permit; Lena has not opted out → governance gate APPROVES
- 3 WITHDRAW — Lena opts out in the wallet → PatientConsent marked withdrawn (auditable, timestamped) → re-run gate → DENIED
- All synthetic: 127 patients · 5,300+ nodes · AlphaKlinik Berlin · PharmaCo Research AG · MedReg DE

The screenshot displays the 'EHDS Compliance Overview' dashboard. At the top, it shows the 'Governance' tab and a navigation bar with 'space', 'Explore', 'Exchange', 'Manage', and 'Docs'. Below the title, it states 'HDAB approval chain · EHDS Art. 45–53' and provides links for 'EEHRx Profiles' and 'Protocol TCK'.

A text box explains the 'EHDS Compliance Matrix' and lists four steps in the approval chain: 1. Access Application, 2. HDAB Review & Approval, 3. Dataset Grant, and 4. Contract.

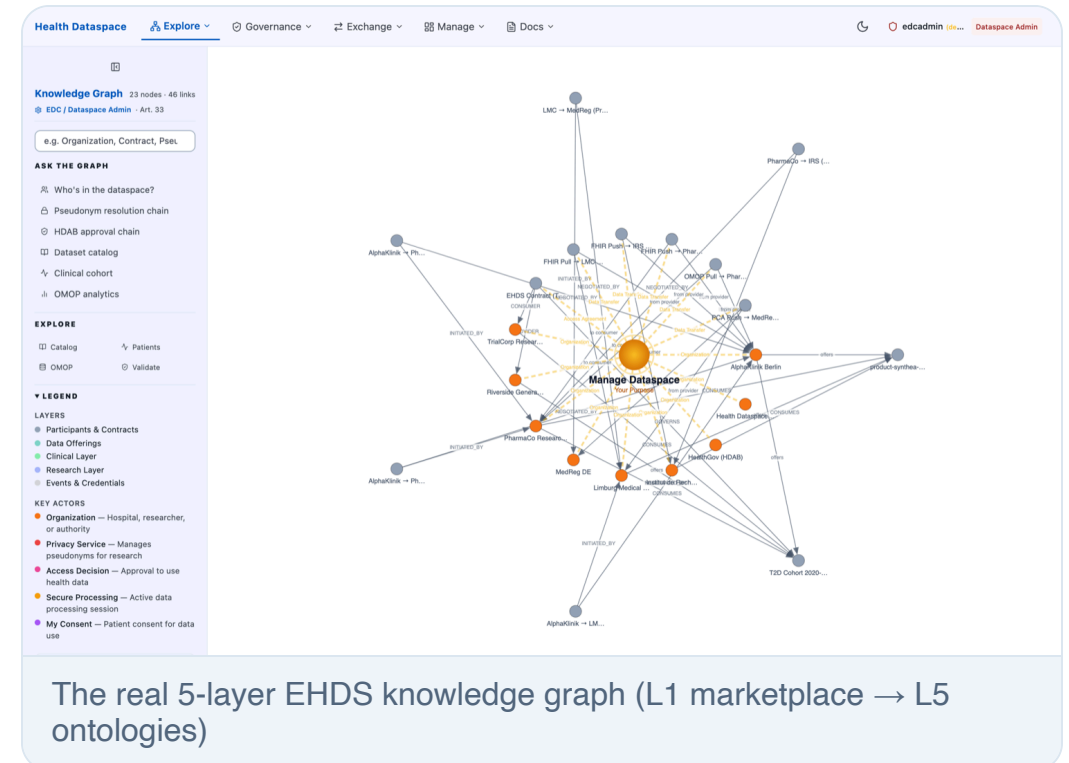
The 'Participant Compliance Matrix' table lists participants and their compliance status:

Participant	Role	Application	HDAB Approval	Dataset	Contract	EHDS Art.	Status
AlphaKlinik Berlin	DATA HOLDER	Pending	—	—	—	—	Pending
HealthGov (HDAB)	HDAB	—	—	—	—	—	HDAB Authority
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Below the table, a text box states: 'MVHD governance check (PASS/FAIL) — the reliable gate visual for the demo'.

5 Why us: regulation, running

- MVHD: a working EHDS reference stack — DSP 2025-1, HealthDCAT-AP, FHIR R4, OMOP CDM, SNOMED/ICD-10/RxNorm/LOINC
- Keycloak OID4VCI bridge issues the SD-JWT preference attestation into the EUDI wallet today
- HDAB permit flow + DCP v1.0 org trust + ODRL 2.2 policies + contract/transfer chain — built
- EUDI iOS reference wallet in simulator; presentation via the hosted EUDI reference verifier — not a 48h mockup



6 Regulation & privacy-by-design

- eIDAS 2.0 (2024/1183) → the attestation: PID + EAA, selective disclosure, data minimisation
- EHDS (2025/327, Ch. IV) → HDAB permit + opt-out: MedReg DE issues the permit; the citizen's opt-out is honoured at the gate
- GDPR → HDAB assesses the Art. 6 basis; Art. 7(3) withdrawal where consent applies; the EHDS opt-out needs no reason (beyond Art. 21)
- Honest scope: enforced at the access boundary today; cryptographic status-list (IETF Token Status List, draft) is roadmap

VISUAL

Article-to-feature mapping: each regulation block wired by an arrow to the exact built feature that satisfies it (eIDAS → EAA / selective disclosure · EHDS Ch. IV → HDAB permit + opt-out honoured at the gate · GDPR Art. 6 basis assessed by HDAB / Art. 7(3) withdrawal). A side ribbon marks built vs roadmap so the honesty is visible, not hidden in the notes.

7 Impact + the ask

- Impact: the trust gap that slows EHDS secondary use — addressed by a citizen-held, verifiable, withdrawable opt-out/consent channel
- Primary ask: shortlist us for EUDI ON — we'll run a live grant → govern → withdraw in Berlin
- Co-develop: standardise the preference-attestation schema (purpose · scope · opt-out) as a reusable EHDS EAA — with HDABs in the loop
- SPRIND/BSI input: treat the in-wallet opt-out as a first-class verifiable trust primitive

VISUAL

Closing: freeze-frame of the wallet opt-out action with the governance panel flipped to DENIED behind it; tagline large across the bottom. QR to the live MVHD portal + repo. A 3-step chevron: Shortlist → EUDI ON live demo → SPRIND/BSI-guided pilot with a fictional HDAB on synthetic cohorts.

8 What's next

- Dynamic, purpose-bound issuance: citizen-selected scope per study (today: one fixed schema, demonstrated)
- Cryptographic revocation: adopt the IETF Token Status List (draft) so verifiers learn opt-out without a server round-trip
- Self-hosted OID4VP verifier path + SPRIND-sandbox PID conformance (not yet achieved)
- Pilot: a fictional HDAB on synthetic cohorts → a reusable pattern any EHDS dataspace adopts

VISUAL

Roadmap horizon graphic in three lanes — Now (built: SD-JWT attestation, HDAB gate, auditable in-graph withdrawal) → Next (status-list revocation, dynamic scope selection, self-hosted verifier) → Pilot (HDAB trial on synthetic cohorts, schema standardisation). Each item tagged built / next / pilot so the jury sees an honest, fundable progression.